

CHAPTER I

INTRODUCTION

Time series analysis is a powerful statistical technique used to analyze and interpret data points collected at successive time intervals. It helps identify underlying trends, seasonal patterns, and irregular fluctuations, making it essential for forecasting in various domains, including finance, meteorology, and transportation. A time series consists of sequential observations where each data point is linked to a specific time stamp. As highlighted by Christopher Chatfield, “a time series is a sequence of observations measured at successive points or over successive periods”. Traditional forecasting methods, such as the Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA), have been widely used due to their ability to capture linear relationships and simple seasonality. However, with increasing data complexity, deep learning models like Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Transformers have emerged as advanced tools that can model intricate temporal dependencies. The selection between these methods depends on factors like dataset complexity, forecast horizon, and computational efficiency.

The dataset used in this study focuses on historical air traffic data spanning from 2003 to 2023, covering crucial indicators such as passenger volume, flight frequencies, revenue passenger miles, available seat miles, and load factors. Air traffic is a fundamental component of global transportation, influencing economic growth, trade, and tourism. Accurate forecasting of air traffic trends is essential for optimizing airline operations, efficient airport management, and effective policy-making. By leveraging time series analysis, this study aims to uncover significant trends and patterns within the dataset, enabling better decision-making in the aviation industry. Traditional statistical methods such as ARIMA and SARIMA are useful for modelling straightforward trends and seasonality, yet they struggle with complex nonlinear patterns. On the other hand, deep learning techniques, particularly LSTM and Transformers, excel in capturing long-term dependencies and intricate relationships within large datasets, making them a promising alternative for air traffic forecasting.

Air traffic data is well-suited for time series forecasting due to its sequential nature and strong temporal dependencies. The patterns observed in air traffic, such as seasonal variations in passenger volume, peak travel periods, and economic influences, make it an ideal candidate for predictive modelling. The data exhibits clear trends and periodic fluctuations, which can be effectively captured using time series models. Additionally, external factors such as fuel prices, global events, and government policies introduce variability in air traffic demand, further emphasizing the need for robust forecasting techniques. By applying time series forecasting models, it is possible to analyze past trends and generate accurate predictions, aiding aviation stakeholders in capacity planning, route optimization, and financial decision-making. The combination of statistical and deep learning models allows for better

adaptation to the dynamic nature of air traffic, ensuring improved forecasting accuracy and reliability.

The key objectives of this study are to evaluate the performance of traditional statistical models (ARIMA, SARIMA) in forecasting air traffic trends, to assess the effectiveness of deep learning models (LSTM, GRU, Transformer) in capturing complex temporal patterns, to develop and test hybrid forecasting models that integrate statistical and deep learning techniques, to compare the forecasting accuracy and computational efficiency of statistical, deep learning, and hybrid models, to provide practical recommendations for airlines, airports, and policymakers on selecting appropriate forecasting methods.

This report is structured as follows: In chapter II the methodology, including dataset selection, model implementation, and evaluation metrics are discussed. In chapter III the experimental results, comparing different forecasting models in terms of accuracy and efficiency of the air-traffic data are presented. Finally, in chapter IV the key findings of the statistical analysis, limitations, and suggestions for future research under air-traffic data are summarized.

CHAPTER II

METHODOLOGY

The monthly air traffic dataset used in this project is taken from Kaggle website. The dataset comprises historical air traffic records, including passenger counts and flight movements used to analyze trends in travel demand and operations. Key features include Date (time index), domestic passengers (captures seasonal domestic trends), international passengers (reflects global travel demand), and total flights (indicates operational capacity).

The dataset contains 249 rows and 17 columns, with no missing or duplicate values. This chapter presents the methodology for air traffic patterns to analyze trends, seasonality, and forecast future demands. It covers traditional statistical methods (ARIMA, SARIMA), deep learning models (LSTM, GRU, Transformer), and hybrid models (e.g., SARIMA + LSTM) for forecasting. Model evaluation is done using metrics like RMSE, MAE, and MAPE. This data supports model training and evaluation, helping forecast air travel demand, identifies seasonality, optimize scheduling, and improve airline decision-making.

2.1 Data Description

The dataset utilized in this study comprises monthly historical air traffic records, originally consisting of 17 columns. However, for the purpose of focused analysis and efficient modeling, only five key variables were selected. They are as follows:

Year	The year of observation (eg., 2003,2023)
Month	The month of observation (eg., 1=January, 12=December)
Dom_Pax	Number of domestic passengers
Int_Pax	Number of international passengers
Pax	Total number of passengers (Dom_Pax + Int_Pax)

2.2 Data Processing

Data preprocessing ensures quality and reliable forecasting. Since the dataset has no missing values, imputation isn't needed. Stationarity is tested using ADF and KPSS; if non-stationary, differencing is applied. This is crucial for ARIMA and SARIMA models. A time-based 80-20 train-test split is used, and expanding window cross-validation improves adaptability. These steps help create a well-structured dataset for accurate and robust air traffic forecasting.

To produce accurate forecasts, the time series must be stationary, meaning it has a constant mean and variance over time. Stationarity is tested using statistical methods like the Augmented Dickey-Fuller (ADF) test and KPSS test. If the series is non-stationary, differencing is applied:

$$y'_t = y_t - y_{t-1}$$

If first-order differencing does not suffice, second-order differencing is used:

$$y''_t = y'_t - y'_{t-1}$$

2.3 Traditional Statistical Methods

2.3.1 Autoregressive Integrated Moving Average (ARIMA)

The Autoregressive Integrated Moving Average (ARIMA) model is a widely used statistical approach for univariate time series forecasting, particularly effective for linear trends. It is defined as ARIMA (p, d, q), where,

AR(p) – an auto regressive model of order p

An auto regressive model represents a type of random process. The output of the model is linearly dependent on its own previous value i.e., some number of lagged data points or the number of past observations. Auto-regressive model of order p can be written as,

$$y'_t = c + \phi_1 y'_{t-1} + \dots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

where ε_t is white noise, p is number of lag observations in the model.

I(d) – Integration of order of differencing

Integrated means the differencing step to generate stationary Time Series data, i.e., removing the seasonal and trend components where 'd' represents number of degree of differencing

MA(q) - Moving Average of order q

Moving Average model in which the output is linearly dependent on the current and various past observations of a stochastic term. A moving average model uses past forecast errors in a regression-like model.

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$$

where ε_t - white noise, q is the order of moving average.

Together these three parts make up the AR-I-MA model. ARIMA(p,d,q) is a linear model originating from the Autoregressive model AR(p), the moving average

model MA(q) and thus the combination of the two AR(p) and MA(q) is the ARIMA (p,d,q)

The optimal values for p, d, and q are selected using Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots. The best model is chosen by minimizing the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

The ARIMA model combines Autoregressive (AR), Integrated (I), and Moving Average (MA) components:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

where,

- y_t is the predicted value at time t
- c is a constant
- ϕ_t are AR coefficients
- θ_j are MA coefficients
- ε_t is a white noise error term

The ARIMA model is effective for forecasting non-stationary time series with trends, offering simplicity and interpretability. However, it struggles with seasonal patterns. To address this, SARIMA extends ARIMA by including seasonal components, allowing it to capture both trend and seasonal variations in air traffic, resulting in more accurate and reliable forecasts.

Despite its limitations, ARIMA remains a strong baseline model for air traffic forecasting, providing reliable short-term predictions for passenger volumes and flight trends based on historical data.

2.3.2 Seasonal Autoregressive Integrated Moving Average (SARIMA)

The Seasonal ARIMA (SARIMA) model is an extension of the ARIMA model, designed to better handle time series data with seasonal patterns. SARIMA is particularly useful for forecasting when data exhibit regular, predictable changes that recur every calendar cycle, such as daily, monthly, or quarterly fluctuations (in short, seasonality). The model is capable of capturing both the trends and seasonality in data to make accurate predictions. To break down each component in SARIMA (p,d,q) (P,D,Q)m. Where,

AR (Autoregressive), p : Number of past observations influencing the present

I (Integrated), d : Number of times the data is differenced to achieve stationarity

MA (Moving Average), q : Number of past error terms used for forecasting.

P : order of seasonal autoregression

D : degree of seasonal differencing

Q : order of seasonal moving average

m : The length of the seasonal period (e.g., 12 for monthly data with an annual cycle).

Autocorrelation Function (ACF) is used to determine the MA (q) and Seasonal MA (Q) components and Partial Autocorrelation Function (PACF) helps identify the AR (p) and Seasonal AR (P) components.

The general SARIMA model combines non-seasonal and seasonal components and is mathematically expressed as:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \sum_{m=1}^P \Phi_m Y_{t-mS} + \sum_{n=1}^Q \Theta_n \varepsilon_{t-nS} + \varepsilon_t$$

where,

- Y_t = actual time series value at time t.
- c = constant term.
- ϕ_i = non-seasonal autoregressive coefficients.

- θ_j = non-seasonal moving average coefficients.
- ε_t = white noise error term.
- Φ_m = seasonal autoregressive coefficients.
- Θ_n = seasonal moving average coefficients.
- S = seasonal cycle length.

SARIMA is effective for capturing trends and seasonality in air traffic data (2003–2023) and offers clear interpretability. However, it struggles with nonlinear patterns, multivariate inputs, and complex dynamics. Deep learning methods like LSTM and Transformers handle these challenges better by modeling nonlinear relationships, long-term dependencies, and external factors - making them more suitable for modern air traffic forecasting.

2.4 Deep Learning Methods

Artificial Neural Network:

Artificial Neural Networks (ANNs) consist of an input layer, one or more hidden layers, and an output layer. Each node in the network is connected and has associated weights and a threshold. A node activates and passes data to the next layer only if its output exceeds the threshold. Inputs are multiplied by their respective weights, summed, and adjusted with a bias. The ANN model is expressed as:

$$\sum_{i=1}^m w_i x_i + \text{bias} = w_1 x_1 + w_2 x_2 + \text{bias}$$

where, w_i = weight assigned to the variable

x_i = input layer

bias = extreme limit (is a constant added to influence the output)

The output of the artificial neural network is,

$$f(x) = \begin{cases} 1 & \text{if } \sum w_1 x_1 + b \geq 0 \\ 0 & \text{if } \sum w_1 x_1 + b < 0 \end{cases}$$

Artificial Neural Networks (ANNs) are powerful models used for various prediction tasks. Among the ANN, Recurrent Neural Networks (RNNs) is well-suited for sequential data, as they retain memory of previous inputs, making them ideal for time series forecasting. Therefore in this project, RNNs is used to forecast air traffic data due to their ability to capture temporal patterns and dependencies. However, standard RNNs suffer from the vanishing gradient problem, making it difficult to capture long-term dependencies.

LSTM networks address this using input, forget, and output gates to manage memory effectively. GRU offers a simpler alternative by combining gates, reducing complexity while preserving performance. Both LSTM and GRU are widely used for tasks involving long-range dependencies in time series and other sequential data.

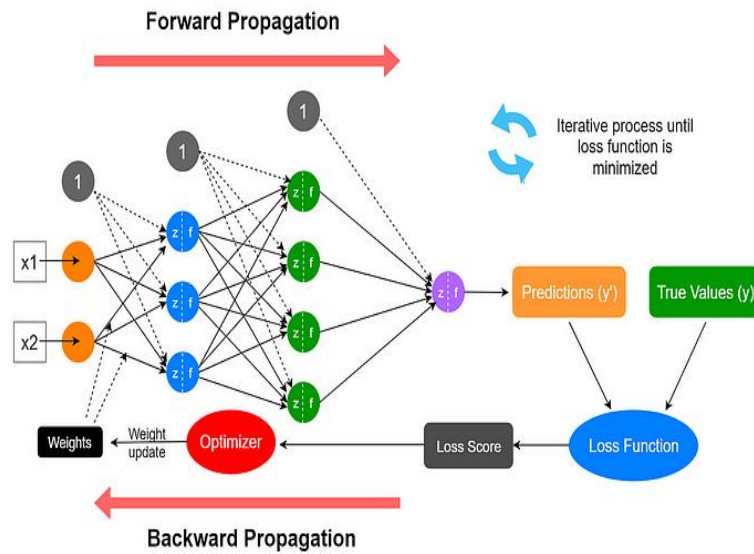


Fig 2.1.Flow Chart for Recurrent Neural Network

2.4.1 Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) designed to process and predicts sequential data by addressing the limitations of standard RNNs, particularly the vanishing gradient problem. Unlike traditional RNNs, which struggle to maintain long-term dependencies due to exponential decay in gradients, LSTMs introduce a memory cell that allows the network to retain relevant information over extended time steps. This makes LSTMs highly effective for tasks such as time series forecasting, speech recognition, machine translation, and anomaly detection, where understanding long-term dependencies is crucial.

LSTM Architecture and Mathematical Formulation

Long Short-Term Memory (LSTM) networks are designed to effectively capture long-range dependencies in sequential data by using a specialized gating mechanism. The architecture includes three primary gates- forget gate, input gate, and output gate - that control the flow of information through the memory cell. At each time step t , the network processes the current input x_t , the previous hidden state h_{t-1} , and the previous cell state C_{t-1}

The forget gate decides which parts of the previous memory to discard and is computed as $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$ where W_f is weight matrix, b_f is bias, h_{t-1} is previous hidden state, x_t is current input, σ is sigmoid activation function, which scales values between 0 and 1. A value close to 0 results in forgetting past information, while a value close to 1 retains it.

The input gate determines how much new information to store, while a candidate memory content $\tilde{C}_t$ is generated as

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Finally, the updated hidden state is calculated $ash_t = o_t * \tanh(C_t)$

This hidden state h_t is then passed to the next time step, allowing information to persist across the sequence. By implementing these gating mechanisms, LSTMs can effectively retain long-range dependencies, making them highly suitable for sequential tasks. However, their step-by-step processing limits parallelization, reducing training efficiency on modern hardware compared to architectures like transformers.

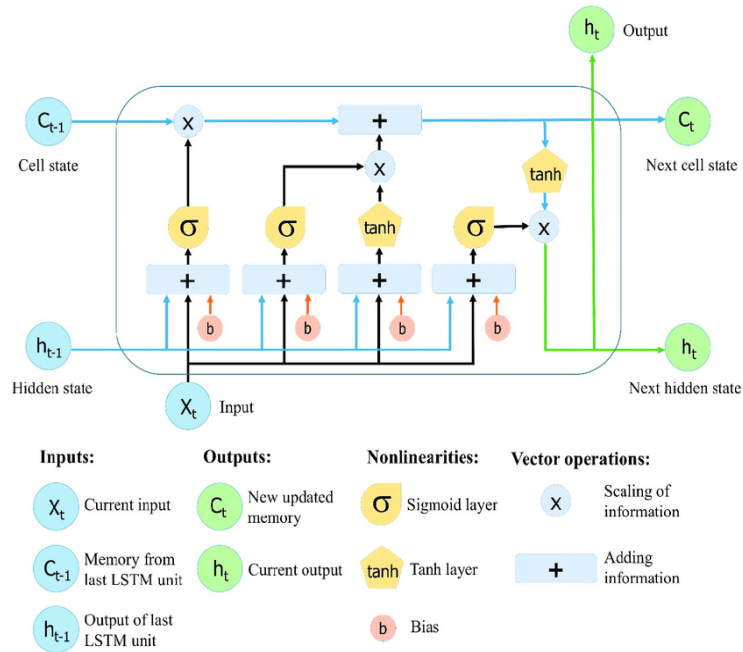


Fig 2.2. Flow Chart for Long Short-Term Memory

GRUs offers a simpler, faster, and more efficient alternative to LSTMs for air traffic forecasting. They maintain similar performance with fewer parameters and lower computational cost, making them suitable for real-time or resource-constrained scenarios. However, they may be slightly less effective than LSTMs when modeling very long-term dependencies.

2.4.2 Gated Recurrent Unit (GRU)

GRU is an advanced form of Recurrent Neural Networks (RNNs) designed to handle sequential data and overcome the vanishing gradient problem. They simplify Long Short-Term Memory (LSTM) networks by combining the forget and input gates into a single update gate, reducing complexity and improving training efficiency. GRUs are well-suited for tasks like speech recognition, machine translation, and financial forecasting, especially where real-time processing and long-range dependencies matter. While they are faster and simpler than LSTMs, they may not always match LSTMs in tasks needing fine memory control.

GRU Architecture and Mathematical Formulation

GRUs consist of two primary gates - reset gate and update gate - which regulate the flow of information and control how past information influences future states. Unlike LSTMs, GRUs don't use a separate cell state instead, they update the hidden state directly. Mathematically, at each time step t , the network processes the current input x_t and the previous hidden state h_{t-1} .

The update gate determines how much of the past information should be retained and how much should be updated with new input. It is computed as $z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z)$, where, W_z is weight matrix, b_z is bias, h_{t-1} is previous hidden state and σ is sigmoid activation function, which scales values between 0 and 1. A value close to 1 retains past information, while a value close to 0 replaces it with new input.

The reset gate controls how much past information should be ignored. It is calculated as $r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$. If r_t is close to 0, the previous hidden state is mostly ignored, allowing the network to focus on new input.

Next, a candidate hidden state is computed using a tan h activation function, which generates a potential new state as $\tilde{h}_t = \tanh(W_h \cdot [r_t * h_{t-1}, x_t] + b_h)$

Finally, the new hidden state is computed as a combination of the previous hidden state and the candidate hidden state, regulated by the update gate as

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

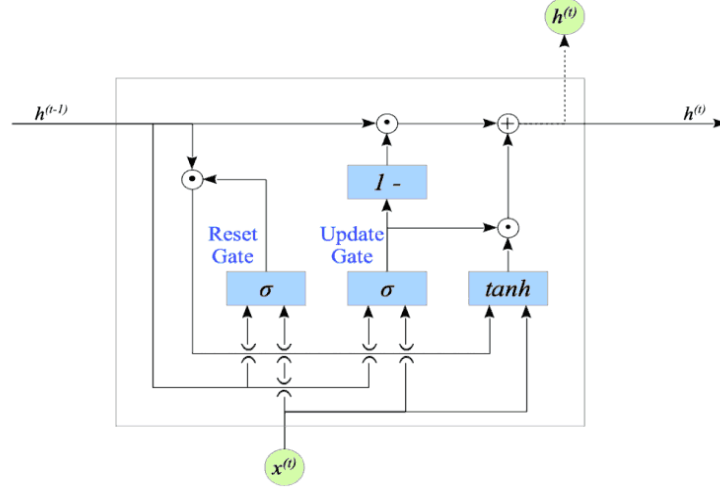


Fig. 2.3. Flow Chart for Gated Recurrent Unit

This formulation allows GRU to selectively retain or discard information at each time step, making them well-suited for sequential tasks while being computationally more efficient than LSTMs. However, because GRUs lack a separate memory cell like LSTMs, they may not always retain long-term dependencies as effectively.

GRU is efficient and fast due to their simplified architecture with fewer parameters, making them ideal for real-time applications. They effectively handle vanishing gradients and capture long-range dependencies. However, they lack a dedicated memory cell, offering less control over memory retention, and may underperform compared to LSTMs on very long sequences. Their merged gating also reduces interpretability.

2.4.3 Transformer Model

The Transformer model, applied in this study for air traffic forecasting, is designed to model complex temporal dependencies without relying on recurrence. It uses a self-attention mechanism to relate every position in a sequence to every other position, allowing it to effectively capture both short and long term patterns. The attention mechanism is based on:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where $Q \in \mathbb{R}^{n \times d_k}$, $K \in \mathbb{R}^{n \times d_k}$, and $V \in \mathbb{R}^{n \times d_k}$ are the query, key, and value matrices respectively, and d_k is the dimensionality of the key vectors.

This operation is extended through multi-head attention, which allows the model to focus on different parts of the sequence simultaneously. The output of the attention layers is passed through position-wise feed-forward networks for non-linear transformation.

Since Transformers process the entire sequence in parallel and lack any inherent notion of order, positional encodings are added to the input embeddings to provide information about the position of each element in the sequence. These encodings are fixed patterns (typically sinusoidal functions) that help the model distinguish between different time steps and learn temporal dependencies effectively.

In this setup, continuous air traffic features (e.g., 11 variables) are first normalized and projected into a higher-dimensional embedding space using a linear transformation. The model is trained using the Mean Squared Error (MSE) loss function:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

To improve training stability and forecasting accuracy, teacher forcing is employed, feeding true values into the decoder. Additionally, causal masking ensures that the model does not use future data points to predict past or present ones, maintaining the integrity of the time series structure.

A hybrid model incorporating deep learning is adopted because traditional models like SARIMA effectively handle linear and seasonal trends but struggle with complex nonlinear patterns and sudden changes. Deep learning models such as LSTM can learn these nonlinear relationships and adapt to shocks, making them powerful for capturing hidden patterns in time series. By combining both approaches, the hybrid model overcomes the limitations of each method individually and achieves more accurate and robust forecasts.

2.5 Hybrid Models

A hybrid model in machine learning and statistical modeling combines multiple algorithms to improve predictive performance by utilizing the strengths of each method while minimizing their weaknesses. In time series forecasting, such hybrid models often merge traditional statistical models like SARIMA which effectively captures linear components such as trend and seasonality with machine learning models like LSTM, which can learn complex nonlinear patterns. The LSTM + SARIMA hybrid model operates in two stages.

First, SARIMA $(p,d,q) \times (P,D,Q,s)$ is applied to model the deterministic (linear and seasonal) aspects of the data. The SARIMA model is expressed as:

$$(1 - \sum_{i=1}^p \phi_i B^i) [(1 - B^s)^D] [(1 - B)^d] Y_t = (1 + \sum_{j=1}^q \theta_j B^j) (1 + \sum_{J=1}^Q \Theta_J B^{Js}) e_t$$

Here, B is the backshift operator, Y_t represents the time series, and e_t is white noise.

In the second stage, the residuals e_t , which contain nonlinear dependencies not captured by SARIMA, are modeled using an LSTM network. LSTM processes these residuals using its gated architecture to learn and forecast the nonlinear patterns.

The LSTM employs a set of gating mechanisms to decide what past information to retain, what new information to add, and how much of the memory should contribute to the output. This allows it to capture long-term dependencies and complex nonlinear relationships in the residuals. The output from the LSTM is then combined with the SARIMA forecast to produce the final prediction. This approach effectively leverages the strengths of both models - SARIMA's ability to model linear trends and seasonality, and LSTM's strength in learning nonlinear patterns, resulting in improved forecasting accuracy.

The final forecast is obtained by combining SARIMA's prediction with LSTM's residual modeling as $Y_t = Y_t^{\text{SARIMA}} + Y_t^{\text{LSTM}}$

This hybrid model offers improved prediction accuracy, making it suitable for applications like air traffic forecasting, stock market prediction, and energy consumption modeling. It balances statistical interpretability with deep learning flexibility. However, it has higher computational complexity, requires careful parameter tuning, and may overfit if not properly optimized. Despite these challenges, the LSTM+ SARIMA hybrid model remains a powerful approach for forecasting time series data that exhibits both linear and nonlinear characteristics.

2.6 Model Evaluation Measures

Evaluation metrics are tools used to assess and analyze the performance of time series forecasting models, serving as key criteria for measuring model performance. Evaluation metrics for time series forecasting include:

Mean Squared Error (MSE): This measures the average squared difference between predicted and actual values, reflecting the overall error between the predictions and actual outcomes. It is calculated as follows:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE): This is the square root of MSE, assigning higher weights to larger errors and emphasizing the stability of the prediction results. It is calculated as follows:

$$\text{RMSE}(y, \hat{y}) = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Mean Absolute Error (MAE): This represents the mean absolute difference between predicted and actual values, reducing the influence of outliers. It is calculated as follows:

$$\text{MAE}(y, \hat{y}) = \frac{1}{n} \sum_{i=1}^n (|y_i - \hat{y}_i|)$$

Mean Absolute Percentage Error (MAPE): This metric considers the relative magnitude of the actual values, avoiding the cancellation effect of positive and negative errors. It is calculated as follows:

$$\text{MAPE}(y, \hat{y}) = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

CHAPTER III

PREDICTION USING TRADITIONAL AND DL MODELS

This chapter presents the results obtained from various forecasting models applied to air traffic data. It includes model performance evaluation, comparative analysis, error analysis, and discussion of key findings. This work results based on the total number of passengers of air traffic data.

3.1 Descriptive Statistics for Passenger Counts

Table 3.1: Descriptive Statistics for Passenger Counts

Metric	Count	Mean	Std. Dev	Min	Max
Dom_Pax	249	55,209,710	10,440,550	2,877,290	75,378,160
Int_Pax	249	7,392,209	2,020,273	136,609	12,432,620
Pax	249	62,601,920	12,292,110	3,013,899	87,810,770

The above Table 3.1 revealed that the average domestic passenger count is significantly higher than the international passenger count averaging to 7.39 million, but with relatively higher variability, likely due to external factors such as travel restrictions, economic conditions, or seasonality. The total passenger count reaches an

average of 62.6 million, with extreme values ranging from 3 million to 87.8 million, due to the COVID-19 pandemic in the year 2020.

3.1.1 Time Plot for Air Traffic



Fig. 3.1. Time Plot for Air Traffic data

The above Fig. 3.1 shows air traffic trends over time. Passenger numbers exhibit steady growth but drop sharply around 2020 due to COVID-19, followed by gradual recovery. The patterns indicate seasonality and long-term trends, requiring further analysis for forecasting accuracy.

3.1.3 Decomposition of Total Passengers into its Components

The below graphs 3.2 represents the decomposition of a time series into its components, the original time series data shows an overall increasing trend over time. A sharp drop is visible around 2020, due to the COVID-19 pandemic affecting air travel. Post-2020, the series appears to recover but with some fluctuations. Next the trend component shows the overall trend is upward, indicating long-term growth. There is a significant drop around 2020, followed by a recovery. Before 2020, minor dips in the trend suggest periods of slower growth. Then the seasonal component shows clear repeating patterns are observed, indicating seasonality.

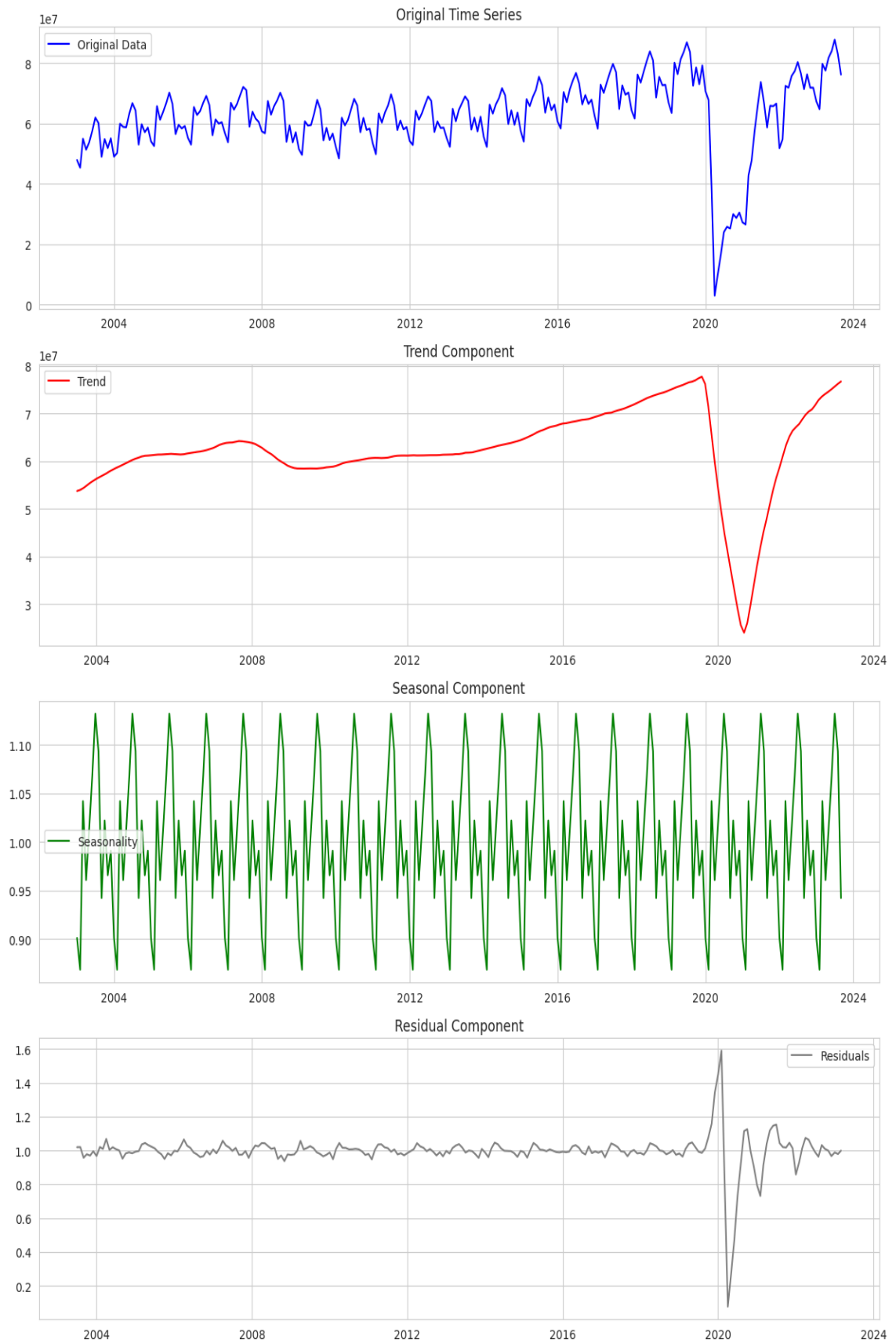


Fig. 3.2. Decomposition of a time series into its components

The seasonal variations remain consistent over the years. The amplitude of seasonal fluctuations is stable, implying predictable periodic behavior. The last graph in the above Fig. 3.2 is the residual component shows that the residuals are mostly stable except for a major anomaly around 2020. This suggests that the model captures most of the structure, but unexpected shocks of COVID-19 create large deviations.

3.1.4 Bar graph for Seasonality

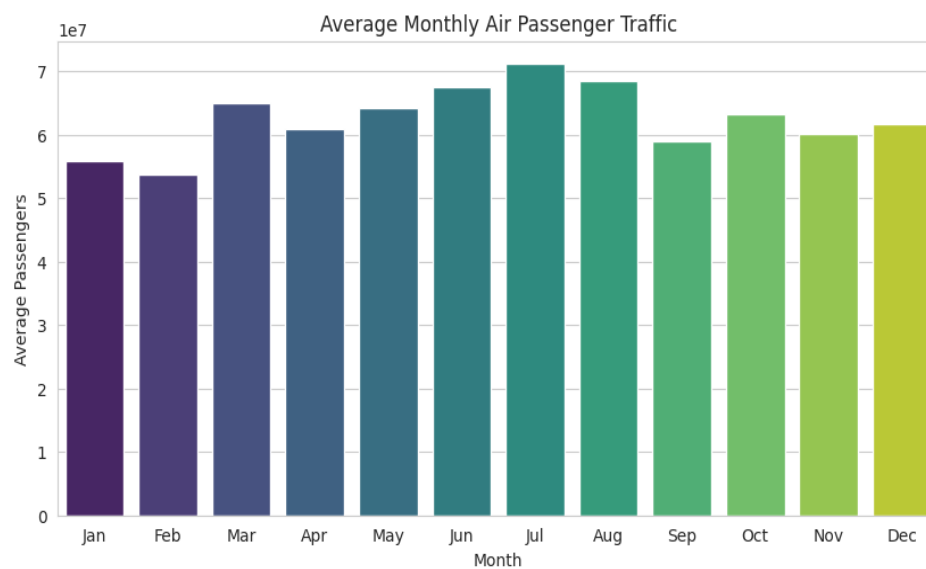


Fig. 3.3. Bar graph for Seasonality

The Fig. 3.3 shows highest passengers in July & August due to summer vacations and lowest in January & February due to post-holiday dip. March-June are Moderate months and October-December show steady demand. It shows clear seasonality, suitable for forecasting trends.

3.1.5 Rolling Mean and Standard Deviation of Total Passengers

ADF Statistic: -3.40045180193928

P-value': 0.010943479350997296

Critical Values: {'1%': -3.4584868856997004, '5%': -2.873918902362675}.

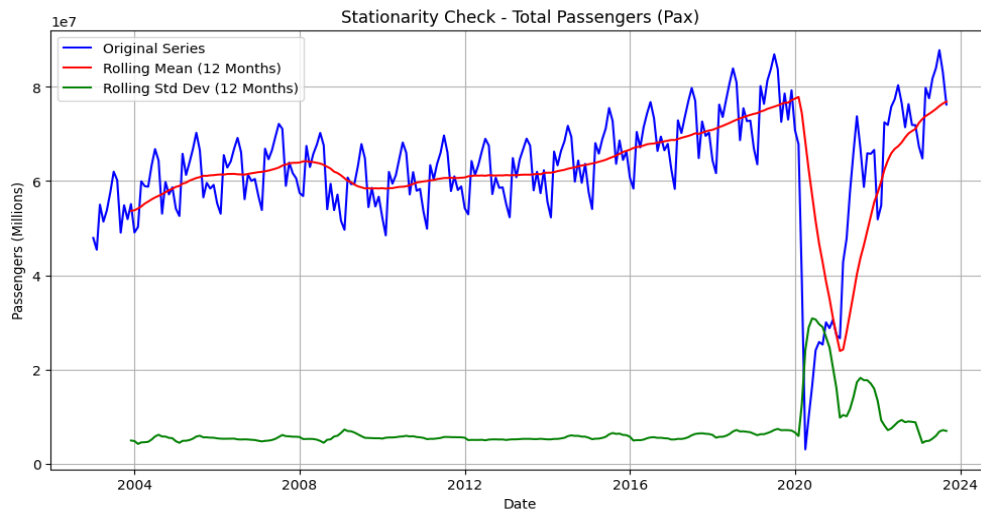


Fig. 3.4. Rolling Mean & Std Dev – Total Passengers

The above Fig. 3.4 shows that the time series is stationary at the 5% significance level, but not at the 1% level. Since the ADF Statistic (-3.400) is less than the 5% critical value (-2.874) and the p-value (0.0109) is less than 0.05, we reject the null hypothesis of the ADF test at the 5% significance level.

3.1.6 ACF and PACF Plot

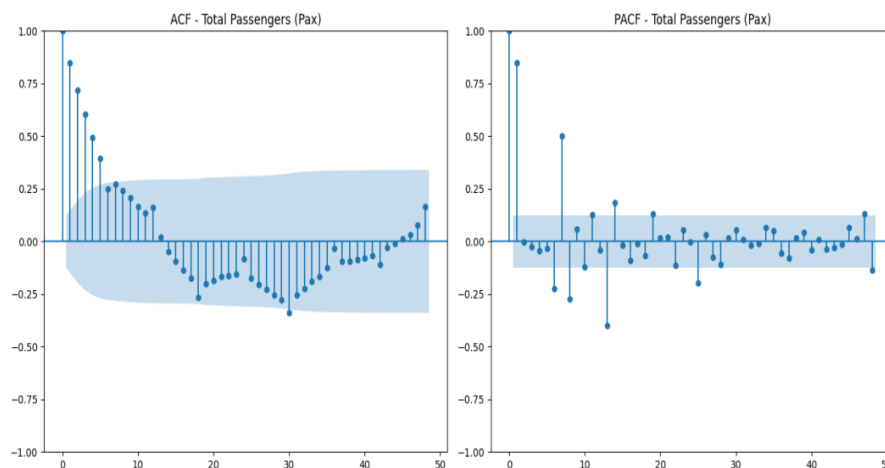


Fig. 3.5. ACF and PACF Plot

The ACF and PACF plots in figure 3.5 for Total Passengers suggest that the time series exhibits autocorrelation and may follow an ARIMA process. The ACF

shows a slow decay, indicating non-seasonal MA components, while the PACF cuts off sharply after lag 1, suggesting a potential AR(1) component. This pattern generally suggests the use of an ARIMA $(1, d, q)$ model, where the differencing order d is determined based on stationarity tests (e.g., the ADF test, which indicates stationarity). A preliminary model to consider is ARIMA $(1, 0, 1)$. Further model selection is be guided by criteria such as AIC and BIC, as well as residual diagnostics. The optimal model will typically exhibit the lowest AIC/BIC values.

3.1.7 Differenced Time Series Data

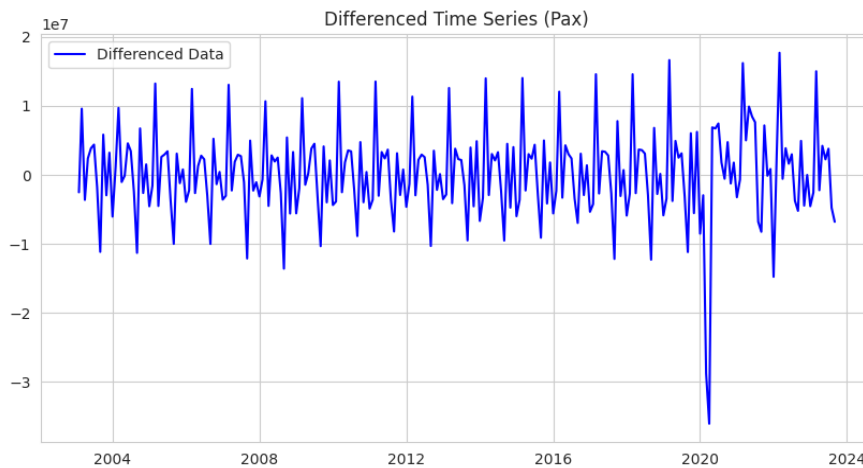


Fig. 3.6. Differenced Time Series Data

ADF Statistic (After Differencing): -4.978826137487458

P-value: 2.4437232456925508e-05.

The differenced time series plot for Total Passengers in Fig. 3.6 appears visually more stationary with constant mean and variance over time, aside from the sharp drop around 2020 (likely due to the COVID-19 pandemic). The ADF test statistic of -4.98 with a p-value of 2.44e-05 confirms this, strongly indicating stationarity after differencing. Therefore, the time series is now suitable for ARIMA modeling with $d = 1$ (i.e., one order of differencing).

3.2 Autoregressive Integrated Moving Average (ARIMA)

3.2.1 ARIMA (1,1,1) Actual vs Fitted

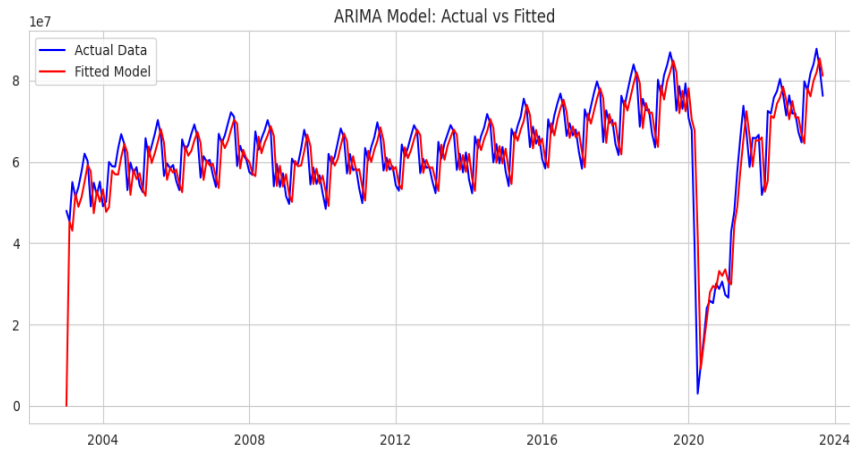


Fig. 3.7. ARIMA (1,1,1) Model Actual vs Fitted

After evaluating various combinations of (p, d, q) , the selected ARIMA model is ARIMA (1, 1, 1). This ARIMA (1,1,1) model in figure 3.7 closely follows the actual data, indicating a good fit. The model captures both seasonality and trends well. Before 2020, the model accurately tracks the actual values with minimal deviations. The periodic fluctuations (seasonality) are well captured by the model. A sharp drop is observed in 2020 due to the COVID-19 impact on air travel. The ARIMA model struggles slightly with capturing the steep decline and subsequent recovery, but it still follows the trend. This suggests that ARIMA may not be optimal for handling sudden shocks or anomalies. The model captures the recovery well, though minor deviations exist. The fit remains strong, implying ARIMA adapts well to post-crisis conditions.

3.2.2 Forecast using ARIMA (1, 1, 1)

The model predicts a slight downward trend in air traffic for the next 12 months. The forecast (dashed orange line) shows a decline instead of continued growth, suggesting the model expects a slowdown. However, the model does not

anticipate further strong growth, possibly due to recent fluctuations in air traffic. The forecast does not fully capture the seasonal peaks and troughs seen in previous years.

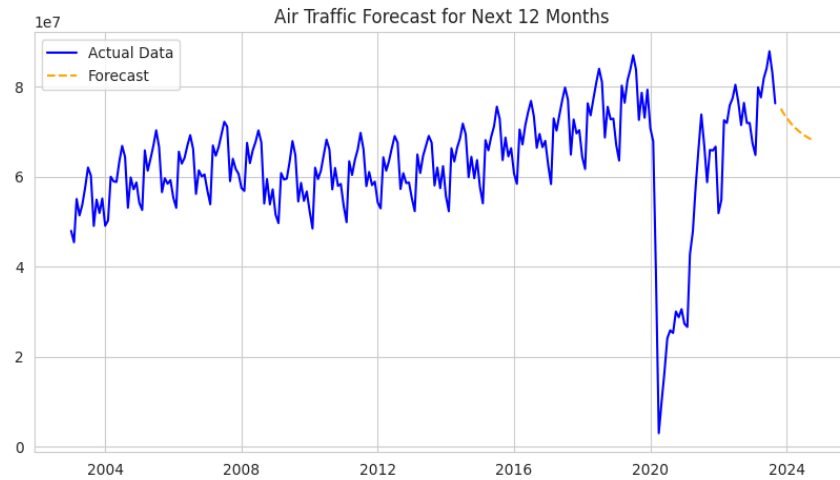


Fig. 3.8. Forecast for Next 12 Months using ARIMA

3.2.3 ARIMA Summary

Table 3.2: Model Evaluation Metrics For ARIMA (1, 1, 1)

Metric	Value
Dependent Variable	Pax
No. of Observations	249
Model	ARIMA(1, 1, 1)
Log Likelihood	-4243.124
AIC	8492.247
BIC	8502.788
HQIC	8496.490
Sample Period	01-01-2003 to 09-01-2023

Table 3.3: Residual Diagnostics of ARIMA model

Test	Value	P-Value
Ljung-Box (L1) (Q)	0.31	0.57
Jarque-Bera (JB)	448.43	0.00
Heteroskedasticity (H)	2.18	0.00
Skew	-1.00	-
Kurtosis	9.28	-

The ARIMA (1, 1, 1) model captures the overall trend and seasonality quite well, with no residual autocorrelation. However, non-normality and heteroscedasticity in residuals suggest potential improvement

The above Table 3.2 shows that the ARIMA (1, 1, 1) model has AIC (8492.247) & BIC (8502.788) values. These values indicate model complexity. Lower AIC/BIC is better. The values are relatively high, suggesting the model could be further optimized.

Ljung-Box Test (Q-statistic = 0.31, $p = 0.57$) has no significant autocorrelation in residuals shows model captures dependencies well.

Jarque-Bera Test (JB = 448.43, $p < 0.001$) residuals are non-normally distributed (high kurtosis, -1.00 skew). It suggests the model does not fully capture the data distribution. Heteroskedasticity ($H = 2.18$, $p < 0.001$) indicates variance instability, possibly due to external shocks (e.g., COVID-19).

3.3 Seasonal ARIMA (SARIMA)

3.3.1 SARIMA Model for Actual vs Predicted Data

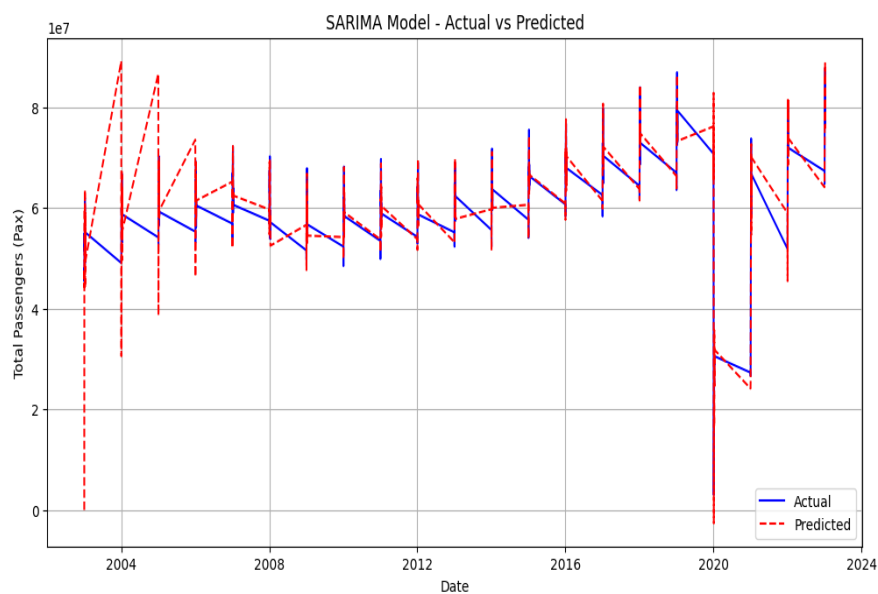


Fig. 3.9. SARIMA Model for Actual Vs Predicted Data

The SARIMA model captures seasonality but shows significant deviations in early years and around extreme events (e.g., 2020 downturn). Predictions seem to lag slightly behind actual values. The high variance in early years suggests instability in the model's initial estimates.

Table 3.4: Model Evaluation Metrics for SARIMA

Dep. Variable:	Pax
Model	SARIMAX(0, 1, 2)x(0, 1, 2, 12)
No. Observations:	249
Log Likelihood	-3498.276
AIC	7006.552
BIC	7023.264
HQIC	7013.309

Table 3.4 infers that SARIMA model is better model fit than ARIMA with AIC = 7006.552 vs. previous 8492.247 implies that it captures both short-term (MA terms) and seasonal effects (MA seasonal terms at 12 and 24 lags).

Table 3.5: SARIMA Model Coefficients

	Coef	std err	Z	P> z	[0.025	0.975]
ma.L1	0.4646	0.048	9.778	0.000	0.371	0.558
ma.L2	0.0780	0.046	1.683	0.092	-0.013	0.169
ma.S.L12	-0.8692	0.071	-12.243	0.000	-1.008	-0.730
ma.S.L24	0.2188	0.062	3.501	0.000	0.096	0.341
sigma2	2.942e+13	1.34e-15	2.2e+28	0.000	2.94e+13	2.94e+13

MA (1) = 0.4646 (strong short-term memory).

Seasonal MA (12) = -0.8692 (strong seasonal dependence).

Seasonal MA (24) = 0.2188 (longer-term seasonal impact)

Table 3.6: Residual Diagnostics for SARIMA

Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	23308.46
Prob(Q):	0.94	Prob(JB):	0.00
Heteroskedasticity (H):	9.91	Skew:	-4.77
Prob(H) (two-sided):	0.00	Kurtosis:	53.85

The above Table 3.6 infers no significant autocorrelation in residuals (Ljung-Box Q-test, $p = 0.94$) implies model is well-specified. Heteroskedasticity is very high ($H = 9.91$, $p < 0.05$) implies variance is unstable. Residuals are highly skewed (-4.77) and leptokurtic (Kurtosis = 53.85) implies extreme values exist due to COVID-19 drop.

3.3.2 Residual of SARIMA Model

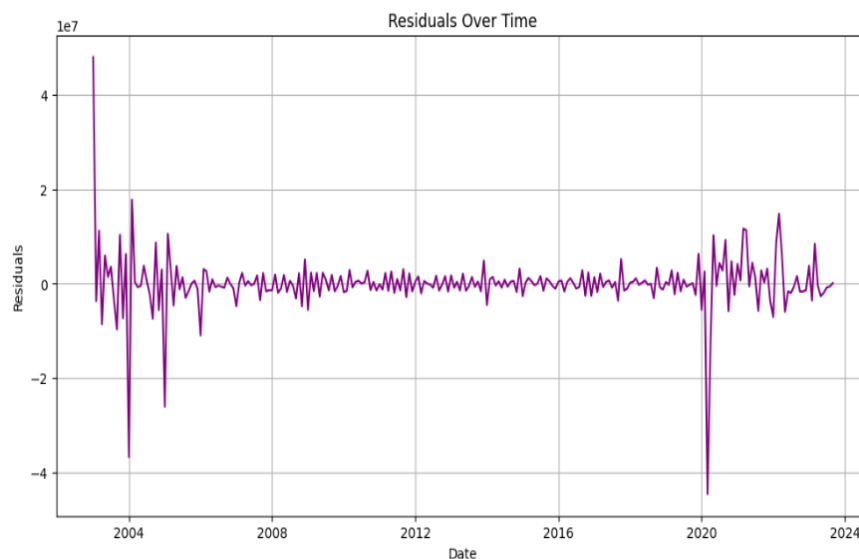


Fig. 3.10. Residual of SARIMA Model

The above Fig. 3.10 shows that the residuals are relatively stable and centered on zero for the majority of the timeline. Noticeable spikes in residuals: Early 2000s: High variability, possibly due to data irregularities or initial instability. Around 2020: Large negative and positive spikes, likely due to the COVID-19 pandemic's impact on air traffic trends.

From 2006 to 2019, residuals are minimal and consistent, indicating good model fit during this period. The SARIMA model performs well during normal conditions, with residuals hovering close to zero. However, extreme deviations around 2020 reflect the model's limitations in handling unforeseen shocks like the pandemic. These periods of increased volatility suggest that external events significantly affected the forecasting accuracy.

3.3.3 Autocorrelation Function of Residuals

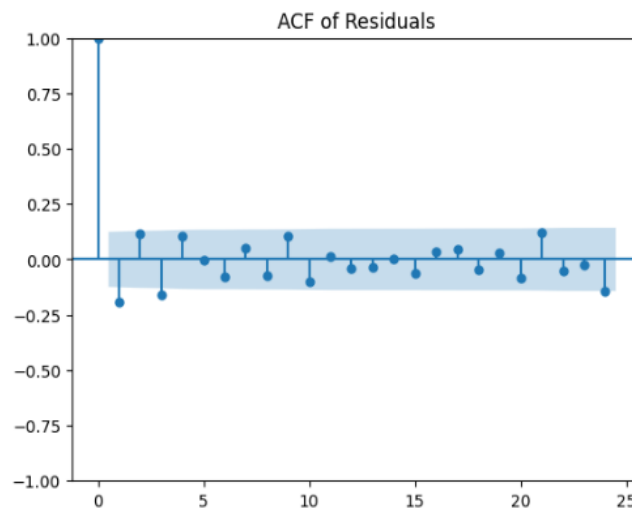


Fig. 3.11. Autocorrelation Function of Residuals

In the above Fig. 3.11, lag 0 shows perfect correlation (as expected). All other autocorrelation spikes lie within the 95% confidence band (shaded blue region). There are no significant spikes beyond the confidence interval, suggesting no autocorrelation in residuals. The residuals appear to be white noise, indicating that the

SARIMA model has effectively captured the underlying structure (trend and seasonality) of the time series. This is a strong diagnostic sign of a well-fitted model.

3.3.4 Forecast Using SARIMA Model

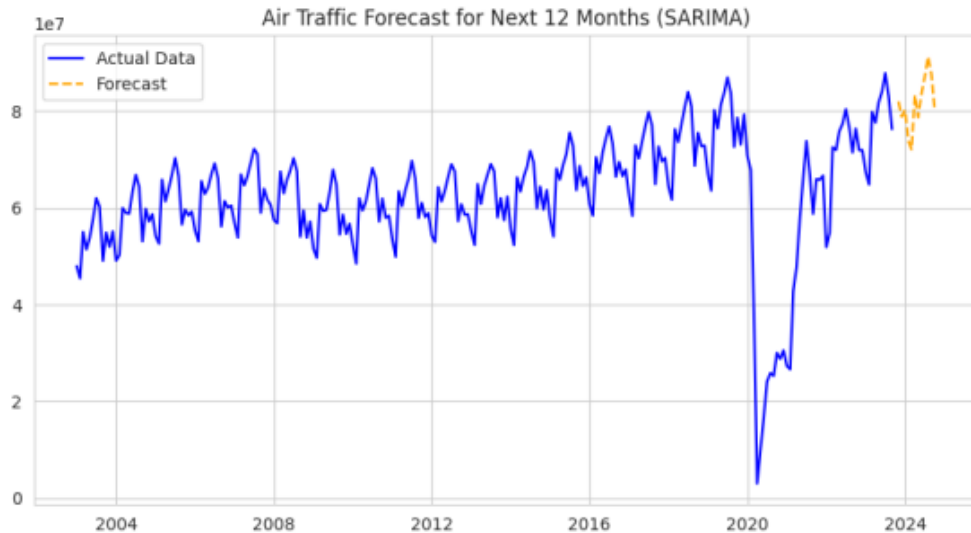


Fig. 3.12. Forecast using SARIMA

The SARIMA model in figure 3.12 successfully captures the seasonal pattern in the historical data. The forecast (dotted line) suggests a gradual recovery after the 2020 dip, aligning with pre-pandemic trends. The forecast appears stable but uncertainty increases towards 2024, visible through fluctuations in the predicted values. This could indicate that the model might need additional external factors (e.g., economic data, airline policies) for better accuracy.

3.3.5 SARIMA Model Evaluation Metrics:

MAE : 3194480.814658776

RMSE : 6560579.557344072

MAPE : 7.966547392751823 %

The SARIMA model performs well, as indicated by a low MAE (3.19M) and RMSE (6.56M). The MAPE of 7.97% suggests that the model is making reasonably accurate predictions with an average deviation of ~8%. The forecasted trend follows historical patterns, suggesting that seasonality and long-term trends are well captured. There is some uncertainty in future forecasts, as shown by the variance in the predicted values. The model handles the COVID-19 dip (2020) but might still be slightly influenced by it. The stability of the forecast suggests that SARIMA has learned from post-2020 recovery trends.

3.4 Long short – Term Memory (LSTM)

3.4.1 LSTM for Actual vs Predicted

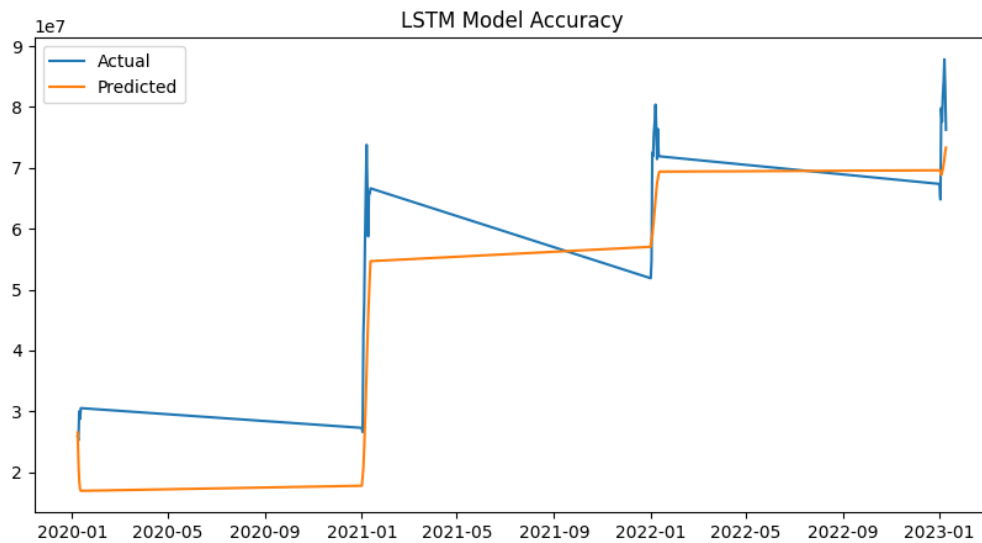


Fig. 3.13. LSTM for Actual vs Predicted

The LSTM model shown in figure 3.13 captures overall trends but struggles with sharp changes and sudden spikes. The model fails to predict extreme fluctuations, especially the sharp drop and recovery around 2020-2021. The predicted values are significantly lower than actual values. This suggests that the LSTM is not fully learning the seasonal and abrupt shifts in air traffic data.

3.4.2 Forecast using LSTM

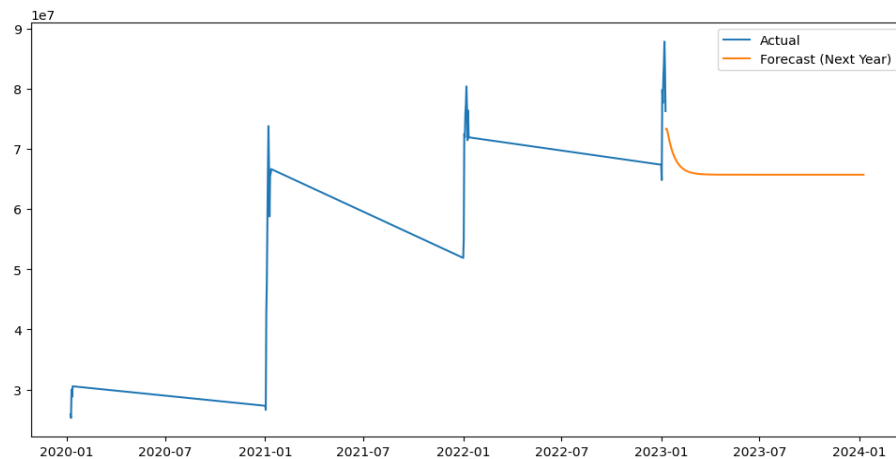


Fig 3.14. Forecast using LSTM

The LSTM model predicts a flattened trend for the next year, suggesting it expects stable passenger traffic instead of a continued seasonal variation. The predicted values appear to converge too quickly, indicating potential under fitting or excessive smoothing.

The actual data shows strong seasonal spikes, but the forecast lacks this periodicity, meaning the model may not be capturing seasonality effectively. This is common in simple smoothing-based models that might struggle with complex seasonality. The sharp drop and recovery in 2020-2021 impact the model's ability to project seasonality into the future. The forecast fails to reflect the periodic trends that were evident before 2020.

3.4.3 LSTM Evaluation Metrics:

MAE : 12475476.37
MSE : 229380069745289.00
RMSE : 15145298.60
MAPE : 21.81%

The MAE of pax is 12.5M and RMSE of pax is 15.1M indicates that the model's predictions deviate significantly from actual values. A high RMSE suggests large variance in errors, meaning the model struggles with certain points, possibly due to extreme fluctuations or outliers. MAPE of pax is 21.81% suggests that, on average, the forecast deviates by 22% from actual values. Generally, a MAPE below 10% is considered good, so this result indicates moderate forecasting accuracy.

3.5 Gated Recurrent Unit (GRU)

3.5.1 GRU Model for Actual vs Predicted

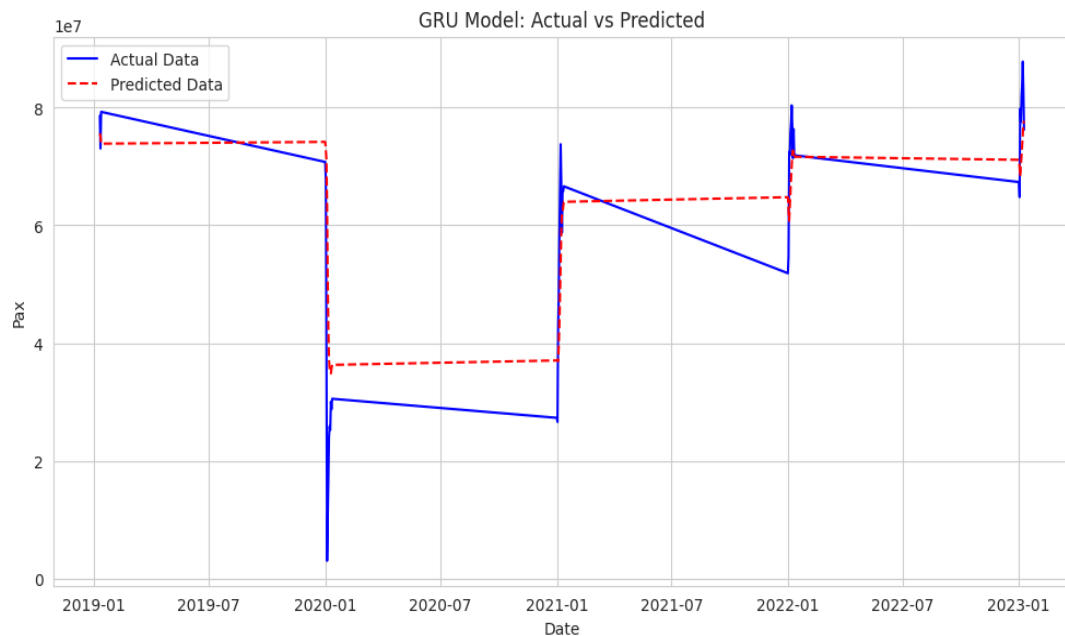


Fig. 3.15. GRU Model for Actual vs Predicted

The Fig. 3.15 reveals key insights into its forecasting performance. The actual data, represented by the blue line, shows significant fluctuations over time, with noticeable seasonal trends and abrupt drops, likely due to external disruptions such as the COVID-19 pandemic. Meanwhile, the predicted values, shown as a red dashed line, appear relatively stable and fail to capture the sharp declines and recoveries in the actual data. This suggests that the GRU model struggles with extreme variations

and may be over-smoothing the predictions, potentially due to inadequate handling of sudden shifts or structural breaks in the time series. The model maintains an overall trend alignment with the actual values but lacks responsiveness to sudden changes

3.5.2 Forecast using GRU

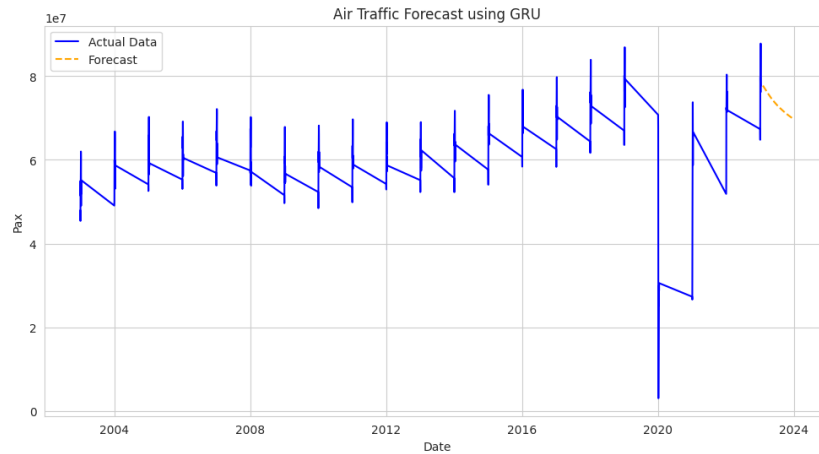


Fig. 3.16. Forecast using GRU

The Fig. 3.16 shows predicted data (dashed red line) is relatively smooth and does not fully capture the abrupt drops and spikes observed in the actual data (solid blue line). This indicates that the GRU model struggles with sudden disruptions or extreme variations. The actual data shows sharp declines and sudden increases, which could be due to external shocks (e.g., COVID-19 pandemic impact on passenger traffic). The GRU model does not fully capture these sudden changes, suggesting a need for incorporating additional external variables or fine-tuning the model to handle shocks better. The predicted values tend to follow a delayed trend compared to actual values, which is a common issue with recurrent models like GRU. This suggests that the model may need better hyper parameter tuning or additional contextual features. Despite missing some sharp transitions, the model still captures the overall long-term trend of passenger traffic reasonably well, indicating that GRU is capable of learning broader patterns but struggles with high-frequency fluctuations.

3.5.3 Evaluation Metrics for GRU

MAE : 10152966.19

RMSE : 14761021.12

MAPE : 69.10%

The error metrics indicate that the GRU model's predictions have significant deviations from the actual values. The mean absolute error (MAE) of 10.15 million suggests that, on average, the model's forecast deviates from the actual values by this amount. The RMSE of 14.76 million highlights even larger errors in some instances, as RMSE penalizes larger deviations more heavily. The MAPE of 69.10% is quite high, indicating that the model struggles with accurate forecasting, likely due to missing seasonality, trend shifts, or external influencing factors. These high error values suggest that the model may need further tuning, additional features, or a hybrid approach to improve accuracy.

3.6 Transformer Model

3.6.1 Forecast using Transformer Model

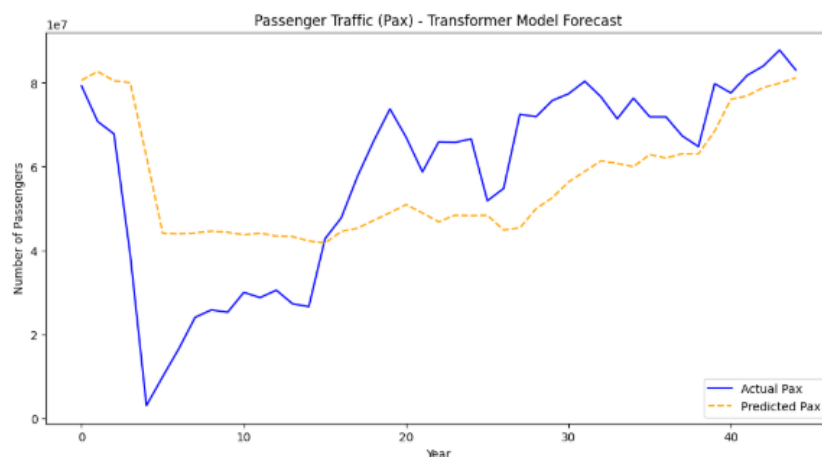


Fig. 3.17. Forecast using Transformer Model

The model in Fig. 3.17 captures the general trend. The predicted values (dashed yellow line) follow the general pattern of the actual values (solid blue line),

indicating that the transformer model has successfully learned the underlying trend in the data. The model tends to underestimate actual values, especially after a significant drop in passenger traffic (early part of the graph). This suggests that the model may struggle with rapid rebounds or sudden shifts in trends. The model does not capture some of the sharp fluctuations seen in the actual passenger data, implying that it may smooth out high-frequency variations. This could indicate a need for further fine-tuning, potentially by incorporating additional features or improving hyperparameters. The transformer model appears promising but might benefit from further training or the inclusion of external factors (such as economic indicators or seasonal effects) to improve predictive accuracy.

3.6.2 Evaluation Metrics for Transformer Model

MAE : 5913918.40

RMSE : 7113226.87

MAPE : 7.91%

The RMSE is approximately 7,113,226.87, indicating that the model's predictions deviate from the actual values by a substantial margin, with greater emphasis on larger errors. The MAE stands at 5,913,918.40, which reflects the average magnitude of errors in the predictions without considering their direction. Meanwhile, the MAPE is 7.91%, suggesting that, on average, the model's forecasts deviate from the actual values by less than 8%, which is generally considered quite well in many forecasting applications. While the high values of RMSE and MAE may be influenced by the scale of the data especially if forecasting variables like total passenger or flight traffic the relatively low MAPE indicates a strong overall performance. Further improvements may be explored through hybrid modeling approaches.

3.7 Hybrid Model

3.7.1 Forecast using Hybrid Model

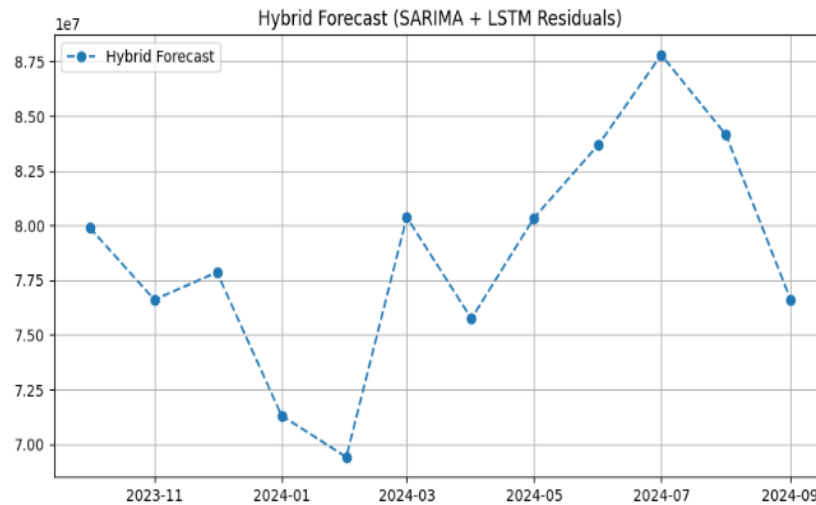


Fig. 3.18. Forecast using Hybrid Model

The Fig. 3.18 captures fluctuations in the data, combining SARIMA's ability to handle linear trends and seasonality with LSTM's capability to capture nonlinear dependencies. The forecasted values exhibit an oscillatory pattern, indicating the presence of cyclic variations. This suggests that the data has periodic fluctuations that the model accounts for. The model captures short-term variations, but some abrupt shifts (such as the sharp increase after 252 and subsequent drop after 258) may indicate potential limitations in forecasting sudden structural changes. While the hybrid model seems to perform well in capturing trends, its accuracy should be evaluated using error metrics like RMSE, MAE, or MAPE to determine its reliability. If residuals (actual - predicted) are small, hybrid model is capturing trends well.

3.7.2 Performance for Hybrid (SARIMA + LSTM):

MAE : 16250718.3069

RMSE : 18133916.9969

R² : 8.5396

The performance metrics for the Hybrid (SARIMA+LSTM) model indicate significant forecasting errors. The mean absolute error (MAE) of 16.25 million suggests that, on average, predictions deviate considerably from actual values, while the Root Mean Squared Error (RMSE) of 18.13 million further highlights that large deviations contributing significantly to overall error. The R^2 score of -8.54 is particularly concerning; as it indicates that the model performs worse than a simple mean-based prediction, suggesting poor generalization and potential model misspecification. These high error values may be due to issues such as improper SARIMA parameter tuning, inadequate LSTM hyper parameter selection, data pre-processing flaws, or a lack of seasonality adjustments.

CHAPTER IV

SUMMARY OF RESULTS

This section presents a comparative analysis of the forecasting models applied to the time series datasets involving air traffic (domestic and international). The performance of each model—ARIMA, SARIMA, LSTM, GRU, Transformer, and Hybrid (SARIMA + LSTM/GRU/Transformer)—was evaluated based on their ability to capture temporal patterns, forecast accuracy, and responsiveness to trend, seasonality, and noise.

Table 4.1: Overall Model Performance Summary

Model	AIC	BIC
ARIMA	8492.25	8502.79
SARIMA	7006.55	7023.26

The Table 4.1 infers SARIMA outperforms ARIMA in terms of both AIC and BIC values, which are significantly lower. Lower AIC and BIC scores indicate a better model fit with less complexity. This suggests that incorporating seasonality in the model (SARIMA) improves predictive performance.

Table 4.2: Model Performance Metrics

Model	MAE	RMSE	MAPE
ARIMA	24362873.25	31985547.84	
SARIMA	3,194,480.81	6,560,579.56	7.97%
LSTM	12,475,476.37	15,145,298.60	21.81%
GRU	10,152,966.19	14,761,021.12	69.10%
Transformer	5,913,918.40	7,113,226.87	7.91%
Hybrid (SARIMA + LSTM)	16,250,718.31	18,133,916.99	Nan%

The Table 4.2 shows SARIMA has best overall performance with the lowest MAE, RMSE, and a very low MAPE (7.97%), indicating accurate and stable forecasting in traditional methods. Transformer also performs well, with low error values and a slightly better MAPE (7.91%) than SARIMA, suggesting it handles seasonality and trends effectively in deep learning methods.

GRU has a moderate RMSE but an extremely high MAPE (69.10%), indicating poor relative accuracy, especially when predicting smaller values. LSTM performs better than GRU but still shows higher errors and MAPE (21.81%), suggesting less accurate forecasting. ARIMA has very high MAE and RMSE, making it less effective than SARIMA and Transformer. Hybrid (SARIMA + LSTM) unexpectedly shows worse performance than SARIMA alone, with very high error values and a missing MAPE, indicating possible integration or data issues in the hybrid model.

SARIMA is the best traditional statistical model due to its ability to handle seasonality effectively. Among deep learning models, Transformer performs best. Hybrid models don't always guarantee better results, as seen with the SARIMA + LSTM combination. This supports the claim that model choice must align with data characteristics, especially when seasonality and trend coexist.

Future research in air traffic forecasting can explore advanced hybrid models that combine statistical and deep learning approaches for improved accuracy. Transformer-based architectures like Informer and Autoformer offer promising results for long-term forecasts. Incorporating external factors such as weather, economic indicators, and holidays can enhance model performance. Real-time adaptive models, probabilistic forecasting, and spatio-temporal techniques (e.g., GNNs) are valuable next steps. Additionally, improving model interpretability, integrating forecasting systems into operational tools, and using transfer learning across regions or domains can broaden the practical impact of this work.

REFERNECES

1. Cai, M, Pipattanasomporn, M & S. Rahman (2019). Day-ahead building-level load forecasts using deep learning vs traditional time-series techniques, *Applied Energy*, 236, 1078–10.
2. <https://www.kaggle.com/datasets/yyxian/u-s-airline-traffic-data/data>
3. <https://search.app/DAhxERth6YXK42SZ9>
4. Kong, X, Chen, Z, Liu1, W, Ning, K, Zhang, L, Marier, S. M, Chen, Y. Y & F. Xia (2025). Deep Learning for Time Series Forecasting: a Survey, *International Journal of Machine Learning and Cybernetics*.
5. Ouwehand, P. S, Krieg & R. Willems (2024). Deep learning for time series forecasting and nowcasting, *CBS discussion paper*.
6. Peter J. Richard, B & A. Davis - *Introduction to Time Series and Forecasting*, 3rd edition, Springer.